An Assessment of Production AEB Failure Modes Under Highway-Speed Lateral Vehicle Intrusion

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Abstract—**Production AEB systems are designed and validated exclusively for longitudinal collision geometries, leaving lateral vehicle intrusion at highway speed unaddressed in any current active safety framework. Analysis of 1.54 million German crash records (2019–2024) reveals motorcycle leaving-carriageway scenarios produce 44.1% severe injury rates—2.37× the national baseline—with zero Euro NCAP AEB protocol coverage.**

This study quantifies production AEB response to highway-speed lateral intrusion using CARLA 0.9.16 physics simulation with 100,000-scenario Monte Carlo analysis spanning speed 60–100 km/h, lateral drift 0.3–1.2 m/s, and lane gap 2.5–5.0 m. Robustness was confirmed across three independent random seeds, eight TTC thresholds, and two target vehicle types.

Production architecture produced collisions in 99.57% of scenarios. Sensors detected the threat at $t = 5.011$ s; TTC-based AEB logic did not activate until $t = 8.050$ s, leaving 1.10 s—insufficient at 80 km/h. Corner radar (120° FOV, 50 m) reduced collision rate to 77.42% (-22.15 pp), 10.1× more effective than a side radar threshold upgrade (-2.19 pp). Perfect detection established a physics-limited floor at 46.62%; best-case actuation reached 16.48%. Failure attribution identifies 53.2% of baseline failures as detection-limited and addressable, and 46.6% as physics-constrained and unavoidable under these kinematics.

Findings reveal structural limitations in both sensor coverage and AEB decision logic that current protocols do not address, and provide a parameterised simulation framework for lateral intrusion test standard development.

Index Terms—**Autonomous Emergency Braking (AEB), Advanced Driver Assistance Systems (ADAS), Lateral collision, Sensor architecture, Monte Carlo simulation, Euro NCAP, CARLA simulator, Motorcycle safety, Highway safety, Scenario validation**

I. Introduction

Modern AEB systems are designed and tested exclusively for longitudinal collision scenarios. Euro NCAP protocols validate rear-end, pedestrian, and cyclist detection—but every test case involves threats approaching from ahead. Euro NCAP's October 2025 Lane Departure Collision protocol introduces Emergency Lane Keeping assessment for scenarios where the ego vehicle drifts laterally toward adjacent vehicles, including motorcyclists [3]. However, this protocol tests steering-based correction for the departing vehicle—not braking response from the receiving vehicle when an adjacent vehicle intrudes into its path. AEB response to lateral intrusion from an adjacent vehicle remains unvalidated in any current active safety framework.

This gap has real-world consequences. Analysis of 1.54 million German crash records (2019–2024) [2], [10] shows motorcycle leaving-carriageway events produce 44.1% severe injury rates, 2.37× the national baseline [10]. Within this category, lateral vehicle intrusion—where a departing vehicle crosses into an adjacent vehicle's path—represents a specific unvalidated topology. Whether production AEB architectures can detect and respond to this scenario remains unknown.

We investigated this using CARLA 0.9.16 simulation with 100,000 Monte Carlo scenarios. A motorcycle initiates lateral departure at 80 km/h, drifting at 0.7 m/s across a 3.5 m lane gap toward the ego vehicle. Production sensor architecture was modelled: forward radar (35° FOV, 150 m), side radar (120° FOV, 8 m, parking-spec $<$10 km/h threshold), rear radar (150° FOV, 80 m). Monte Carlo parameters spanned speeds 60–100 km/h, drift rates 0.3–1.2 m/s, gaps 2.5–5.0 m, and offsets ± 10 m. Three counterfactual sensor configurations were tested: improved side radar, corner radar (120° FOV, 50 m), and perfect detection. TTC sensitivity analysis across eight thresholds (1.5–5.0 s) validated results robustness.

Results show production architecture produces collision in 99.6% of scenarios. Sensors detect threats at $t = 5.01$ s, but TTC-based AEB logic does not activate until $t = 8.05$ s—leaving 1.1 seconds insufficient at highway speed. The failure is architectural, not parametric: all eight TTC thresholds tested produced collision outcomes. Corner radar reduces collision rate to 77.4% (-22.1 pp), 10.1× larger improvement than side radar upgrade alone (97.4%, -2.2 pp). Perfect detection establishes a floor at 46.6%, attributing 53.2% of failures to detection limits and 46.6% to physics constraints. Target validation using a passenger car (Toyota Prius) produced identical timing, confirming architecture-level causation within CARLA's geometric sensor model.

This work contributes quantitative evidence of lateral detection gaps through physics-based simulation, isolates sensor versus decision logic bottlenecks via counterfactual analysis, and connects simulation findings to real-world crash severity data. CARLA's geometric model represents an upper bound on detection—real motorcycle radar cross-section would worsen baseline performance. Results identify architectural gaps requiring investigation, not production failure rates. Hardware modifications require validation per ISO 21448.

II. Related Work

A. AEB System Design and Longitudinal Focus

Autonomous Emergency Braking research has concentrated almost exclusively on longitudinal collision geometries. Euro NCAP test protocols validate AEB performance against rear stationary targets, rear moving vehicles, pedestrians, and cyclists—all involving threats along the vehicle's longitudinal axis [4]. NHTSA's FMVSS No. 127 mandate similarly requires AEB systems to detect forward collision threats, with

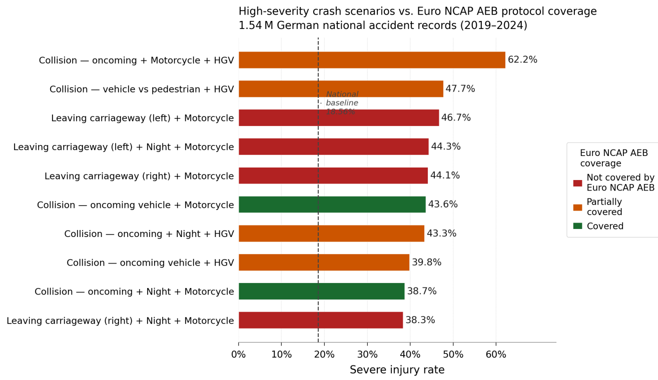

Fig. 1. Phase 1 finding from 1.54 M German crash records: motorcycle leaving-carriageway scenarios show 44.1% severe injury rate, 2.37× national baseline, with zero Euro NCAP AEB coverage.

performance requirements specified up to 62 mph for vehicle-to-vehicle scenarios [7]. Hu et al. investigated AEB key parameters for improving car-to-two-wheeler collision safety using in-depth traffic accident data, demonstrating that TTC threshold calibration for two-wheeler scenarios significantly affects collision avoidance outcomes in longitudinal following situations [12]. Kim et al. proposed AEB simulation methods based on vehicle test data for ADAS accident analysis, again within forward collision geometry [13]. The consistent longitudinal focus across both regulation and research reflects the geometric assumptions embedded in production AEB architectures—and defines the boundary at which this study begins.

B. TTC-Based Decision Logic and Lateral Threat Geometry

Time-to-collision, formalised by Hayward [5], provides the mathematical foundation for AEB activation timing. TTC assumes high closing speeds along the longitudinal axis, producing rapid threshold crossing that enables early warning. This assumption transfers poorly to lateral threat geometries. Behera et al. confirmed explicitly that "the conventional formulation of TTC does not account for the risk of lateral collisions, such as sideswipes, where contact occurs along the sides of the vehicles" [6]. Under lateral drift at 0.70 m/s, TTC declines 2.5× more slowly than in a standard 80 km/h forward approach—creating a fundamental mismatch between the metric's calibration domain and lateral threat kinematics. Extensions to two-dimensional TTC formulations have been proposed [6], but production AEB systems continue to operate on standard one-dimensional TTC logic optimised for longitudinal geometry.

C. Motorcycle Detection and Simulation-Based ADAS Validation

Motorcycle detection presents specific challenges for radar-based ADAS systems due to lower radar cross-section (∼1–3 m^2) compared to passenger cars (∼10–20 m^2) and a narrower lateral profile, making reliable detection harder at range [4]. Existing motorcycle-specific ADAS studies have focused on detection accuracy, lane departure warning, and forward collision avoidance—not on AEB response to lateral intrusion from an adjacent motorcycle into the ego vehicle's path. CARLA has become a standard platform for physics-based ADAS scenario testing, supporting reproducible parametric studies and configurable sensor models [1]. Euro NCAP's 2025 Lane Departure Collision protocol [3] introduced the first formal active safety assessment for car-to-motorcyclist lateral scenarios—testing Emergency Lane Keeping for the departing vehicle. The complementary case—AEB response for the receiving vehicle when an adjacent vehicle intrudes laterally at highway speed—has not been addressed in the published literature. This study fills that specific gap through CARLA-based simulation and large-scale Monte Carlo analysis.

III. Methodology

A. Simulation Environment

Scenarios were implemented in CARLA 0.9.16 [1], an open-source autonomous driving simulator providing deterministic physics and configurable sensor models. Simulations executed on Windows 11 (AMD Ryzen 7 250, NVIDIA 5060 Mobile, 32 GB RAM) using Python 3.10 with NumPy 1.24, Matplotlib 3.7, and SciPy 1.11. Complete dependency specifications are available in the GitHub repository [11].

The Town04 map was selected for its straight highway section, eliminating curve geometry as a confounding variable. Weather was set to clear conditions with fixed midday lighting for consistent sensor performance. Vehicle models were selected from CARLA's blueprint library to represent a mid-size crossover ego vehicle (BMW Grand Tourer), a motorcycle primary threat (Kawasaki Ninja, S1), and a sedan robustness check (Toyota Prius, S2). Fig. 2 shows the CARLA simulation sequence.

Scenarios executed in synchronous mode at 20 Hz. Both vehicles travelled at 80 km/h with 3.50 m lateral separation. After a 5.0 s stable phase, the threat vehicle initiated constant 0.70 m/s lateral drift while maintaining forward velocity. This replicates the kinematic signature of a vehicle losing lateral control—the topology producing 44.1% severe injury rates in Phase 1 crash data [10] with zero Euro NCAP coverage.

B. Sensor Architecture Modelling

Production sensor architecture was modelled using CARLA's ray-cast detection with industry-representative specifications. Table I summarises sensor configurations for baseline and counterfactual architectures. Figs. 3–5 show the geometric coverage analysis and direct visual proof of the detection gap.

Critical modelling constraint: CARLA uses geometric ray-cast visibility without radar cross-section (RCS) modelling, multipath, or clutter. This represents an upper bound on detection capability. Real motorcycle RCS (∼1–3 m^2) would degrade detection relative to cars (∼10–20 m^2), making the reported 99.6% baseline a lower bound on real-world failure rates.

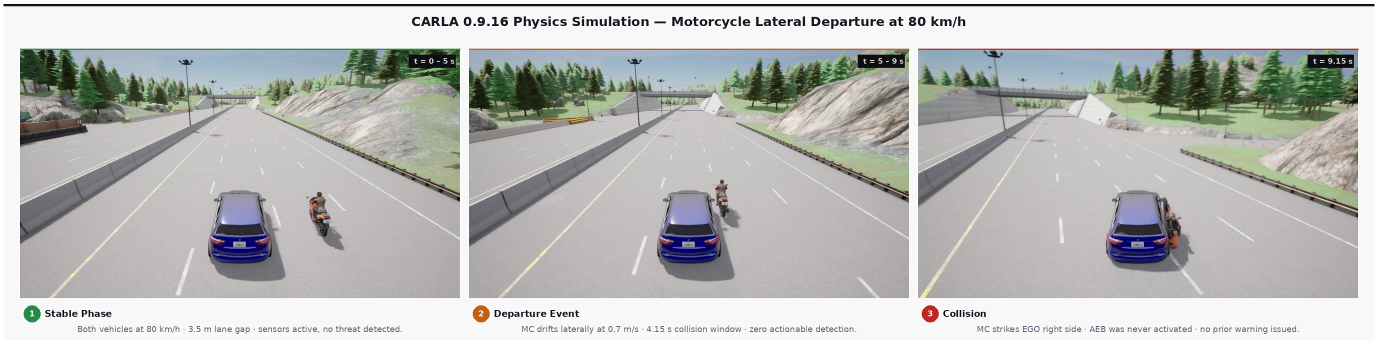

Fig. 2. CARLA 0.9.16 simulation sequence showing lateral departure scenario: stable travel phase ($t = 0$–5 s), lateral departure ($t = 5$–9 s), collision ($t = 9.15$ s) despite target visibility.

TABLE I
SENSOR ARCHITECTURE SPECIFICATIONS

Config	Sensor	FOV	Range
Baseline	Forward radar	35°	150 m
	Side radar	120°	8 m
	Rear radar	150°	80 m
Improved	Side radar	120°	8 m (highway)
Corner	Corner radar	120°	50 m
Perfect	Geometric ideal (upper bound)		

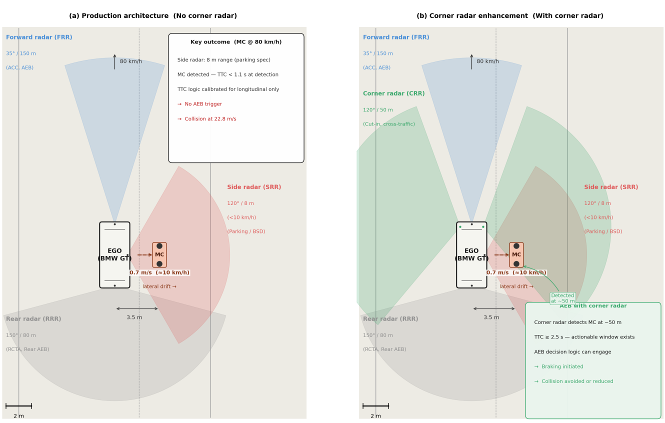

Fig. 3. Production sensor architecture (left) exhibits lateral blind zone from 35° forward FOV and parking-spec side radar; corner radar enhancement (right) restores lateral coverage.

AEB decision logic: TTC activation used TTC = lateral_gap/drift_velocity. AEB triggered when TTC ≤ 3.43 s AND detection = True. Collision occurred when $t_{\text{cross}} < (t_{\text{react}} + v_{\text{ego}}/a_{\max})$, with baseline parameters: $t_{\text{react}} = 1.0$ s, $a_{\max} = 8.0\,\text{m/s}^2$.

C. Kinematic Model and Correction Factor

The mathematical prediction for lateral crossing time is:

$$t_{\text{cross}}^{\text{pred}} = \frac{d_{\text{lat},0}}{v_{\text{drift}}} = \frac{3.50\,\text{m}}{0.70\,\text{m/s}} = 5.0\,\text{s} \tag{1}$$

CARLA measurement yielded a collision at $t = 9.15$ s after departure onset at $t = 5.00$ s, giving an observed duration of 4.15 s. This discrepancy arises because collision detection triggers at a finite lateral separation rather than zero.

The correction factor was derived as a **kinematic threshold ratio**:

$$S_3 = \frac{d_{\text{lat},0} - d_{\text{thresh}}}{d_{\text{lat},0}} = \frac{3.50 - 0.60}{3.50} = 0.83 \tag{2}$$

where $d_{\text{lat},0} = 3.50$ m is the initial lane gap and $d_{\text{thresh}} = 0.60$ m is the physics handoff threshold approximating motorcycle body half-width. Fig. 5 illustrates the kinematic model geometry. Table II shows validation across five drift speeds.

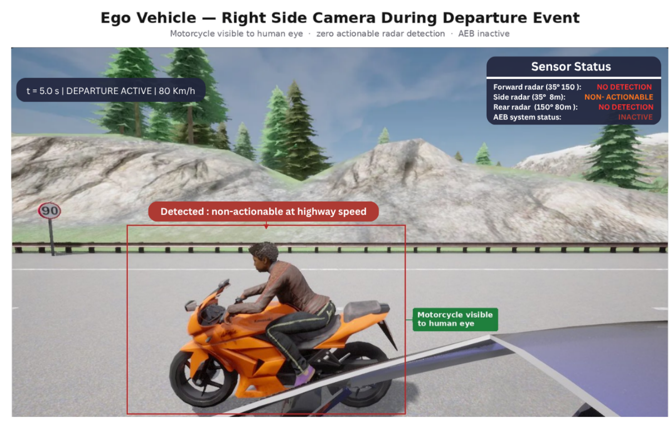

Fig. 4. CARLA frame at $t = 5.0$ s: motorcycle visible but zero actionable detection (forward FOV gap, side radar non-actionable at highway speed).

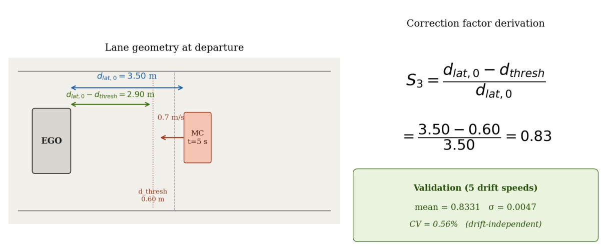

Fig. 5. Kinematic derivation of the correction factor applied to physical lateral reach.

TABLE II
CORRECTION FACTOR VALIDATION ACROSS DRIFT SPEEDS

Drift (m/s)	Pred. (s)	Obs. (s)	Ratio	S_3
0.3	11.67	9.70	0.8314	0.83
0.5	7.00	5.80	0.8286	0.83
0.7	5.00	4.15	0.8300	0.83
0.9	3.89	3.25	0.8357	0.83
1.2	2.92	2.45	0.8400	0.83
Mean	—	—	0.8331	0.83
Std	—	—	0.0047	—

Consistency across drift speeds ($\sigma = 0.0047$, CV $= 0.56\%$) confirms the ratio is drift-independent. The correction was applied globally:

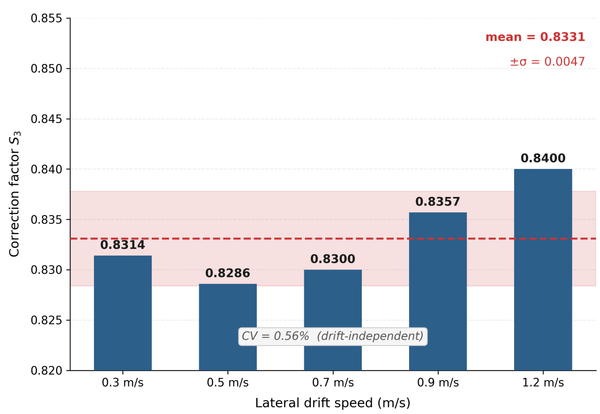

Fig. 6. Correction factor validation: mean $= 0.8331$, $\sigma = 0.0047$ across drift range (0.3–1.2 m/s), confirming drift-independence.

$$t_{\text{cross}} = \frac{d_{\text{lat}}}{v_{\text{drift}}} \cdot S_3 = \frac{d_{\text{lat}}}{v_{\text{drift}}} \cdot 0.83 \tag{3}$$

D. Monte Carlo Framework

Analytical model: Monte Carlo analysis evaluates a closed-form collision model calibrated from CARLA observations, not 100,000 independent CARLA runs. This enables computational tractability (3–5 min on standard hardware) while preserving physics fidelity.

The analytical collision model is defined by three equations. The corrected crossing time is given by (3). The required response time is:

$$t_{\text{respond}} = t_{\text{react}} + \frac{v_{\text{ego}}}{a_{\max}} \tag{4}$$

where t_{react} is the system reaction delay (s), v_{ego} is ego vehicle speed (m/s), and $a_{\max}$ is maximum deceleration (m/s^2). A scenario produces a collision outcome when:

$$t_{\text{cross}} \leq t_{\text{respond}} \quad \vee \quad \delta_{\text{detect}} = 0 \tag{5}$$

where $\delta_{\text{detect}} = 0$ indicates no actionable sensor detection under production specifications.

Parameter distributions: Table III lists the four independent uniform distributions spanning realistic highway conditions. The ± 10 m longitudinal offset range models pure lateral departure geometry only, excluding cut-in scenarios that would artificially inflate AEB viability.

TABLE III
MONTE CARLO PARAMETER SPACE

Param.	Dist.	Range	Rationale
Speed	Stratified uniform	60–100 km/h	60% weight in 70–90 km/h highway band
Drift	Uniform	0.3–1.2 m/s	Gradual drift to rapid loss-of-control
Gap	Uniform	2.5–5.0 m	European highway lane width variation
Offset	Uniform	± 10 m	True lateral geometry (excludes cut-ins)

Execution: NumPy's `default_rng()` with explicit seeding (42, 123, 999) ensured reproducibility. Each seed generated 100,000 independent scenarios. Output: collision rate = proportion of scenarios where AEB intervention fails. Fig. 7 presents multi-seed validation results (std $= 0.020\%$).

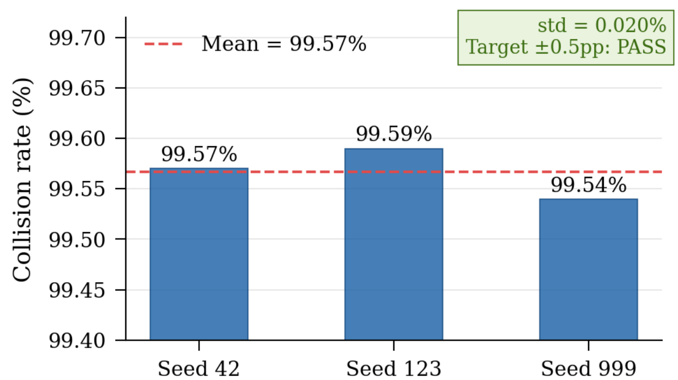

Fig. 7. Multi-seed validation: collision rates 99.57%, 99.59%, 99.54% for seeds 42/123/999 (std $= 0.020\%$), confirming statistical stability.

E. Counterfactual Analysis

Four sensor configurations (Table I) were evaluated across the same 100,000-scenario parameter space to isolate detection versus decision logic contributions. Configuration comparison quantifies:

1) Whether speed threshold or range limits side radar performance.
2) Effectiveness of corner sensor enhancement.
3) The physics-limited floor under perfect detection.

Actuation parameter sweep: Under perfect detection, a 4×3 grid tested reaction times (1.2, 1.0, 0.8, 0.5 s) and deceleration rates (8.0, 9.0, 10.0 m/s^2). The 0.5 s / 10 m/s^2 combination represents idealised best-case performance. Results establish the minimum achievable collision rate under the modelled lateral kinematics at 80 km/h.

F. Robustness Validation

Three independent validation tests confirmed results stability.

TTC sensitivity analysis: Eight activation thresholds (1.5–5.0 s) were tested under baseline sensor configuration to verify collision outcomes are threshold-independent. Fig. 8 presents results.

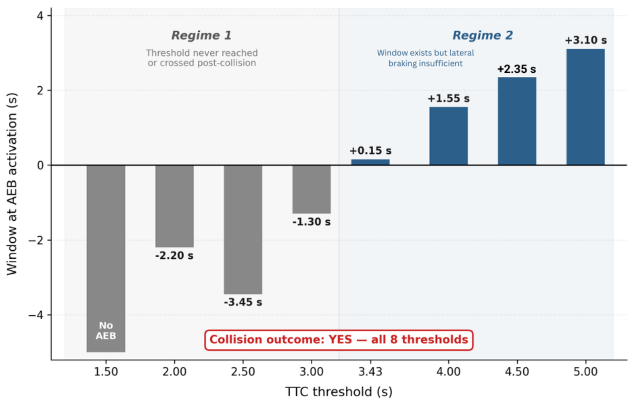

Fig. 8. TTC threshold sensitivity (1.5–5.0 s): collision outcomes in all eight configurations, confirming failure is architectural and TTC-independent.

Note on $\mathrm{TTC}_{\mathrm{lat}}$ *dynamics:* $\mathrm{TTC}_{\mathrm{lat}}$ is defined as:

$$\mathrm{TTC}_{\mathrm{lat}}(t) = \frac{d_{\mathrm{lat}}(t)}{v_{\mathrm{drift}}} \tag{6}$$

By definition, $\mathrm{TTC}_{\mathrm{lat}}$ declines at 1 s per real second. Since lateral drift speed ($v_{\mathrm{drift}} = 0.70$ m/s) is far lower than longitudinal closing speed ($\sim$22 m/s at 80 km/h), $\mathrm{TTC}_{\mathrm{lat}}$ values remain elevated for significantly longer than in equivalent forward collision scenarios—explaining why the threshold is crossed late and leaves insufficient response window.

Multi-seed convergence: Three random seeds (42, 123, 999) generated independent 100,000-scenario samples. Standard deviation across seeds quantifies statistical stability. Target: ± 0.5 pp around the expected 99.6% baseline. Formally:

$$\sigma_{\mathrm{seeds}} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\left(r_i - \bar{r}\right)^2} < 0.5\,\mathrm{pp} \tag{7}$$

where r_i is the collision rate for seed i, $\bar{r}$ is the mean across $N = 3$ seeds. The observed value $\sigma_{\mathrm{seeds}} = 0.020\%$ confirms the target is met by a factor of 25.

Target-type validation: Passenger car (S2) scenarios were executed under identical parameters (80 km/h, 3.5 m gap, 0.7 m/s drift) to test whether timing depends on vehicle morphology or represents an architecture-level failure. Complete dependency specifications are available in the GitHub repository [11]. Fig. 14 compares S1 vs. S2 system-level outcomes.

IV. Results

A. Baseline Performance

Under production sensor architecture, 99.57% of the 100,000 simulated lateral intrusion scenarios resulted in collision. This does not mean the system malfunctioned. What the data shows is that the modelled production architecture produced no actionable AEB response under these conditions. To confirm the finding is statistically stable, the same analysis was repeated with three independent random seeds (42, 123, 999), producing 99.57%, 99.59%, and 99.54% — a standard deviation of 0.020%. The result does not change with the random seed. Fig. 9 shows how outcomes are distributed across the full parameter space.

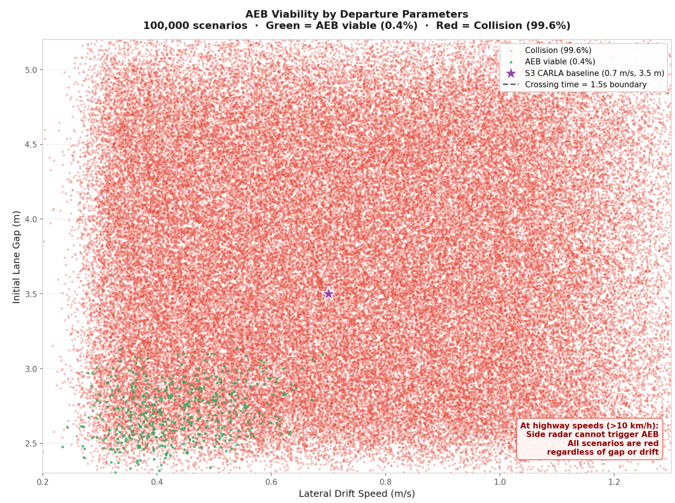

Fig. 9. Parameter space heatmap across drift rate and lane gap: realistic highway conditions fall entirely within the collision zone. The 0.4% AEB-viable scenarios exist only at extreme slow-drift and large-gap boundaries.

B. Detection Window Analysis

The sensors detected the threat immediately. All three production sensors registered the departing motorcycle at $t = 5.011$ s, just 0.011 s after lateral drift began at $t = 5.000$ s. Detection was not the problem. Table IV traces what happened from detection to collision.

TABLE IV
Event Timing Sequence (Baseline Scenario)

Event	Time	TTC remaining
Departure begins	$t = 5.000$ s	5.00 s
Detection onset	$t = 5.011$ s	4.99 s
TTC threshold crossed	$t = 7.950$ s	3.43 s
AEB activation	$t = 8.050$ s	1.10 s
Collision	$t = 9.150$ s	0 s

Detection happened at $t = 5.011$ s. AEB activated at $t = 8.050$ s. Three seconds passed between the two. The reason is how TTC-based decision logic works under lateral kinematics. In a standard forward collision, a vehicle approaching at 80 km/h closing speed causes TTC to drop fast—crossing the activation threshold quickly. In a lateral intrusion at 0.70 m/s drift, TTC declines slowly: from 5.00 s at departure to 3.43 s at $t = 7.95$ s—a 2.95-second decline across 2.95 seconds of real time. The logic waited for a threshold it was calibrated for a completely different geometry. By the time AEB activated at $t = 8.05$ s, only 1.10 s remained. At 80 km/h, a full emergency stop requires at minimum 3.2 s at 10 m/s^2 deceleration. The window was not close. Fig. 10 puts this against Euro NCAP benchmarks.

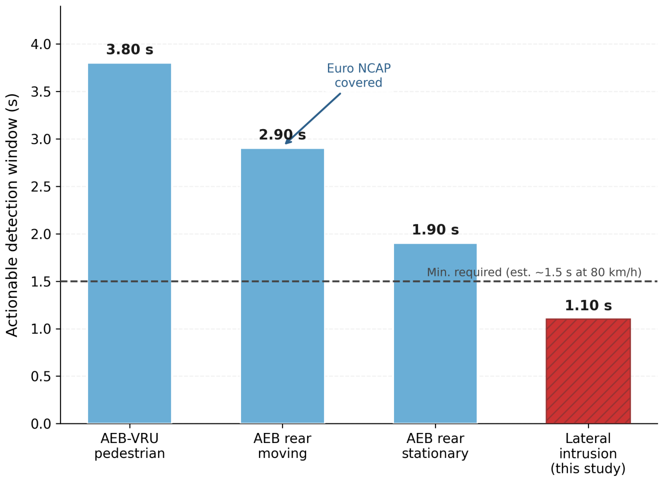

Fig. 10. Response windows: Euro NCAP longitudinal scenarios provide 1.9–3.8 s of actionable time; the lateral intrusion scenario provides 1.10 s—after sensors had already detected the threat 3 seconds earlier.

C. Counterfactual Architecture Results

Four sensor configurations were tested to identify which architectural changes produce meaningful improvement. Table V presents the results.

TABLE V
COUNTERFACTUAL ARCHITECTURE PERFORMANCE

Configuration	Collision rate	Δpp
Baseline (production)	99.57%	—
Improved side radar	97.38%	−2.19
Corner radar (120° FOV/50 m)	77.42%	−22.15
Perfect detection	46.62%	−52.95

Raising the side radar operational threshold from below 10 km/h to highway-capable speed reduced collision rate by 2.19 percentage points. The change is negligible because side radar range is still 8 m. Even a highway-rated side radar cannot detect a lateral threat far enough in advance at these speeds. Range is the bottleneck, not the speed threshold. Adding corner sensors (120° FOV, 50 m range) reduced collision rate by 22.15 percentage points — 10.1× more effective than the side radar upgrade. Corner placement addresses the geometric blind zone created by the 35° forward radar and provides the range needed for early detection. Both factors matter together: wider field of view and longer range. Removing all sensor constraints entirely produced a floor of 46.62%. Even with instant, perfect detection in every scenario, nearly half still end in collision. This is not a sensor limitation — it reflects scenarios where the available departure-to-impact window is geometrically too short for any braking system to prevent collision at 80 km/h. This floor is physics, not hardware. Failure attribution follows directly: 52.95 pp (53.2%) of baseline failures are detection-limited and can be reduced through better sensor architecture. The remaining 46.62% are physics-limited and cannot be avoided through sensor or braking improvements alone under the modelled kinematics. Figs. 11 and 12 present these results visually.

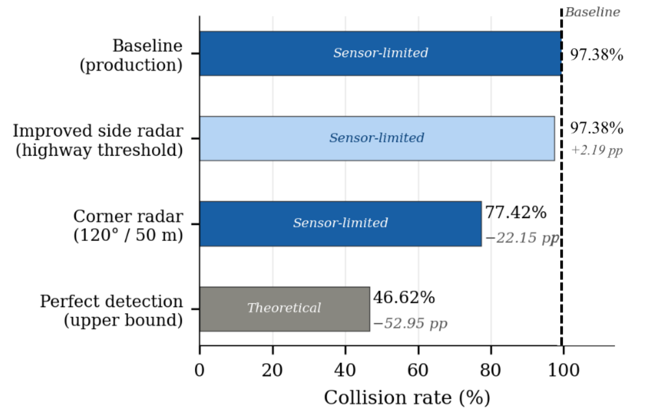

Fig. 11. Counterfactual comparison: corner radar reduces collisions by 22.15 pp—10.1× more than the side radar upgrade (2.19 pp). Range and field-of-view both matter.

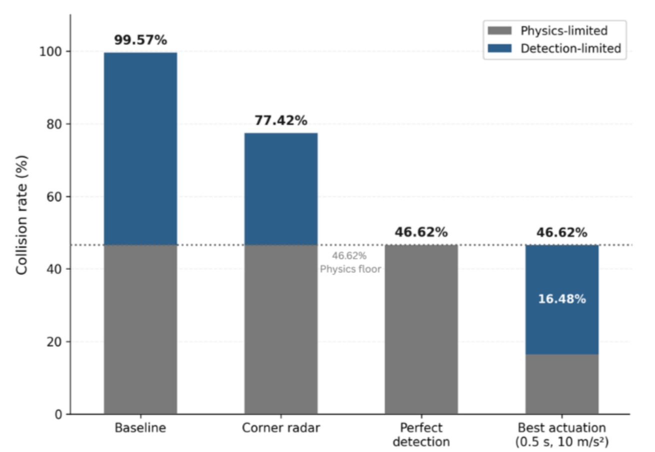

Fig. 12. Failure attribution: 53.2% of collisions are detection-limited and addressable through sensor architecture improvement. The remaining 46.6% are physics-limited and unavoidable under the modelled lateral kinematics at 80 km/h.

D. Actuation Parameter Sweep

With sensor constraints removed (Perfect Detection), braking parameters were varied across 12 combinations to find the minimum collision rate. Table VI shows the full sweep.

The best result—0.5 s reaction time and 10.0 m/s^2 deceleration under perfect detection—was 16.48%. Approximately 1 in 6 scenarios still ends in collision. Longitudinal braking reduces ego vehicle speed but cannot stop the lateral closing motion of the departing threat. Some departure trajectories are geometrically unavoidable through braking alone at these

TABLE VI
ACTUATION SWEEP UNDER PERFECT DETECTION

Reaction (s)	Deceleration (m/s^2)		
	8.0	9.0	10.0
1.2	46.48%	39.65%	33.61%
1.0	42.16%	34.88%	28.64%
0.8	37.61%	29.92%	23.72%
0.5	30.31%	22.64%	**16.48%**

speeds. This confirms the physics-limited floor established in Section III-C. Fig. 13 shows the full parameter sweep.

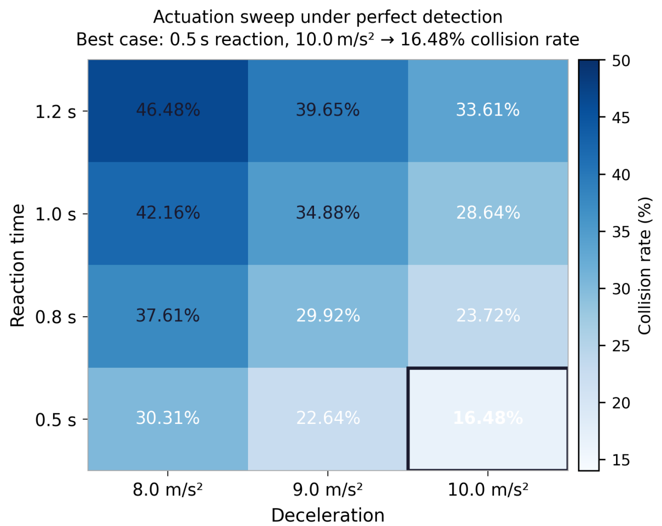

Fig. 13. Actuation sweep under perfect detection: best case (0.5 s, 10 m/s^2) achieves 16.48%. The collision floor persists across all parameter combinations.

E. TTC Threshold Sensitivity

The baseline uses a TTC activation threshold of 3.43 s. Production AEB systems from different manufacturers may use different values. The question is whether the 99.57% finding depends on this specific choice. Eight thresholds from 1.5 s to 5.00 s were tested. All eight produced collision outcomes. Table VII shows the full results.

TABLE VII
TTC THRESHOLD SENSITIVITY (BASELINE ARCHITECTURE)

TTC (s)	AEB fires?	Window (s)	Collision
1.50	No	—	Yes
2.00	Yes	−2.20	Yes
2.50	Yes	−3.45	Yes
3.00	Yes	−1.30	Yes
3.43	Yes	+0.15	Yes
4.00	Yes	+1.55	Yes
4.50	Yes	+2.35	Yes
5.00	Yes	+3.10	Yes

Two regimes emerge and both end in collision, for different reasons. Regime 1 (TTC 1.5–3.0 s): The minimum TTC reached during the entire 4.15 s departure window is 3.34 s. Thresholds below this value are never crossed while the collision is still preventable. For TTC = 1.5 s, AEB never activates at all. For TTC = 2.0, 2.5, and 3.0 s, the threshold is eventually crossed — but only after the collision at $t = 9.15$ s, when CARLA continues running post-impact. The negative window values (−2.20 s, −3.45 s, −1.30 s) mean AEB would become active 1.30 to 3.45 seconds after impact had already occurred. These are not near-misses. They are post-collision events with no physical relevance. Regime 2 (TTC 3.43–5.0 s): AEB activates before collision, with available windows of 0.15 s to 3.10 s. Even the largest window — 3.10 s at TTC = 5.0 s — is insufficient. Longitudinal braking at 80 km/h reduces ego vehicle speed but does not stop lateral closing motion. The collision happens regardless of how early the brakes engage. The 99.57% result does not depend on which threshold is used. The architecture fails across the entire realistic production range.

F. Target-Type Validation

Everything above was tested with motorcycle (Kawasaki Ninja) as the departing vehicle. The same scenario was run with a passenger car (Toyota Prius, S2) under identical conditions: same speed, same gap, same drift to check whether the findings are specific to motorcycles or reflect something deeper about the architecture. Table VIII compares both.

TABLE VIII
MOTORCYCLE VS. PASSENGER CAR TIMING COMPARISON

Metric	Motorcycle (S1)	Car (S2)
Detection onset	5.011 s	5.002 s
AEB activation	8.050 s	8.050 s
Collision	9.150 s	9.150 s
Response window	1.10 s	1.10 s
Outcome	Collision	Collision

The car was detected 0.009 s faster than the motorcycle, its larger geometric cross-section is easier for CARLA's raycast sensor model to see. Beyond that, every critical timing is identical: AEB activated at the same moment, collision occurred at the same moment, response window was the same. The outcome was the same.

This tells us the problem is in how the decision logic responds to lateral kinematics—not in the specific target vehicle. It is an architecture-level finding, not a motorcycle-specific one. One important caveat: CARLA's geometric sensor model does not simulate real radar cross section differences. Real motorcycles (∼1–3 m^2 RCS) are significantly harder to detect than cars (∼10–20 m^2). In a real system, motorcycle detection would be degraded further. The car validation establishes internal simulation consistency—not real-world equivalence between target types.

V. DISCUSSION

A. The Two-Layer Failure

The results isolate a two-layer architectural mismatch within the modelled production configuration. Layer 1 operates at

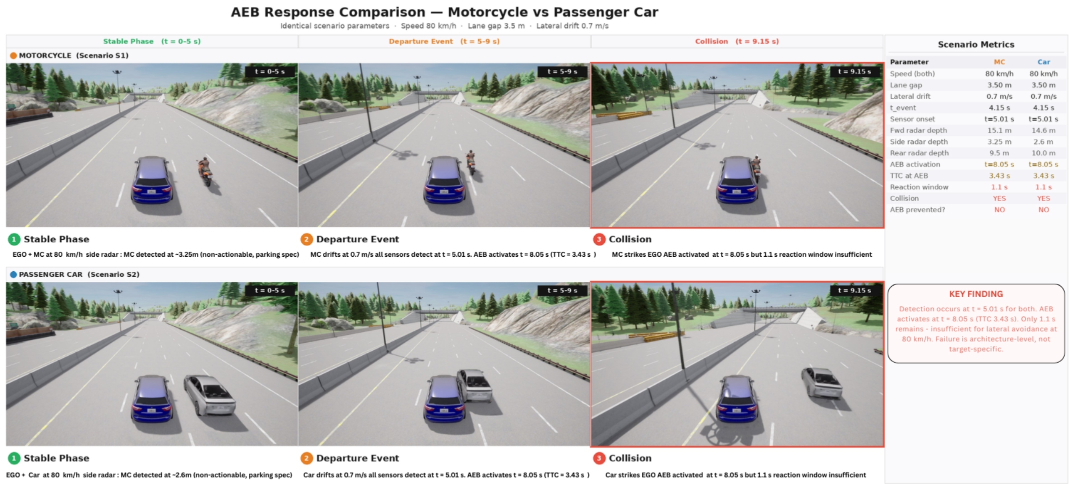

Fig. 14. Target-type comparison: motorcycle (left) and passenger car (right) produce identical AEB timing and collision outcome, confirming architecture-level causation within CARLA's geometric sensor model.

the sensor level; Layer 2 at the decision logic level. Both must be addressed simultaneously — counterfactual results confirm that resolving either layer alone produces insufficient improvement. What the data shows is that the modelled production architecture produced no actionable AEB response under these conditions.

Layer 1 — Sensor coverage gap: The 35° forward radar creates a geometric blind zone for lateral targets. Side radar detects within 8 m but is specified for parking-assist below 10 km/h — non-actionable at highway speed.

Layer 2 — Decision logic mismatch: Sensors detected the threat at $t = 5.011$ s with 4.14 s remaining. Decision logic consumed 3.04 s—73% of that time—before AEB activated, leaving 1.10 s. Physically insufficient at 80 km/h.

Both layers compound each other. Fixing one without the other produces minimal improvement, as the counterfactual results confirm.

B. Why TTC Logic Does Not Transfer to Lateral Threats

Under lateral intrusion at 0.70 m/s drift, TTC declines at 0.395 s per real second—2.5× slower than a standard 80 km/h longitudinal approach (∼1.0 s/s). The same threshold providing 3+ seconds of actionable warning in a forward collision provides only 1.10 s here. Literature confirms this directly: conventional TTC "does not account for the risk of lateral collisions such as sideswipes" [6].

TTC sensitivity analysis confirms the failure is structural. Across eight thresholds from 1.5 s to 5.0 s, collision occurred in every configuration. The minimum TTC reached during the 4.15 s departure window is 3.34 s. Thresholds below 3.34 s are geometrically unreachable before impact—AEB never fires or fires post-collision. For thresholds above 3.34 s, AEB activates with windows of 0.15–3.10 s, but longitudinal braking cannot stop lateral closing motion regardless of activation timing. No threshold prevents this collision. The architecture needs to change, not the parameter.

C. Counterfactual Interpretation

Side radar threshold upgrade (8 m range retained): −2.19 pp. Range is the binding constraint — 8 m is insufficient for early highway-speed warning regardless of threshold setting. A software-only fix to the existing side radar does not work.

Corner radar (120° FOV, 50 m): 22.15 pp—10.1× larger improvement. Both FOV and range are addressed simultaneously. Neither change alone would produce this result. Corner radar sensors are commercially available from automotive Tier-1 suppliers and currently deployed for blind-spot monitoring — not integrated into highway-speed AEB pathways. The hardware exists. The validated integration does not.

At 77.42%, corner radar still leaves a significant collision rate. The 30.80 pp gap between corner radar and perfect detection reflects scenarios where even 50 m coverage cannot provide sufficient warning — departure-to-impact windows too short for that detection range. Corner radar is a meaningful step, not a complete solution.

D. Physics-Limited Floor

Perfect detection produced a 46.62% floor; best-case actuation (0.5 s, 10.0 m/s^2) reached 16.48%. Longitudinal braking reduces ego vehicle speed but cannot stop lateral closing motion. For fast drift rates, small gaps, or close offsets, the departure-to-impact window is too short for braking alone. Complete avoidance likely requires lateral evasive capability beyond longitudinal braking — a direction for future work.

E. Connection to Phase 1 Crash Data

In 99.57% of modelled scenarios, no effective AEB intervention occurs, and impact takes place at near-full relative speed — mean 22.77 m/s. At these speeds, severe injury outcomes are expected, consistent with Phase 1's 44.1% severe injury rate in motorcycle leaving-carriageway events. The connection is directional, not quantitative. Phase 1's leaving-carriageway category includes barrier impacts, rollovers, and other outcomes beyond lateral vehicle intrusion. Phase 2 tests one specific topology within that category and explains architecturally why it produces severe outcomes.

F. Limitations

Sensor modelling: CARLA's geometric ray-cast model is an upper bound on detection. Real motorcycle RCS ($\sim$1–3 m^2) would degrade detection relative to cars ($\sim$10–20 m^2). Real-world failure rates would be equal to or worse than 99.57%. The baseline is conservative.

Analytical Monte Carlo: 100,000 scenarios evaluate a closed-form model calibrated from CARLA observations—not 100,000 independent physics runs. $S_3 = 0.83$ validated across five drift speeds (CV $= 0.56\%$). Gap variation beyond 3.50 m was not independently validated in CARLA. Multi-seed validation confirmed stability (std $= 0.020\%$).

Scenario scope: CARLA physics validation used a straight highway at 80 km/h. Monte Carlo extends analytically to 60–100 km/h. Constant drift, no evasive steering, and linear braking are deliberate scope decisions to isolate the architectural failure mechanism.

Interpretation boundary: 99.57% represents parameter space coverage under modelled conditions—not a real-world crash probability. Production validation and ISO 21448 (SOTIF) [8] compliance verification are required before these findings inform any design or regulatory decision.

G. Implications

Sensor architecture: Corner sensors address both FOV coverage and range simultaneously. The 10.1× improvement ratio identifies this as the hardware direction most worth investigating. The 30.80 pp gap to perfect detection suggests sensor fusion beyond single-modality radar warrants further study.

Decision logic: Conventional TTC does not extend to lateral threat geometry [5], [6]. A framework computing lateral closing speed and time-to-lane-crossing independently from longitudinal TTC would address the 3.04-second delay identified here. This requires new algorithm development — not threshold recalibration.

Validation protocols: Euro NCAP's October 2025 Lane Departure Collision protocol (v1.1) introduces Emergency Lane Keeping assessment for Car-to-Motorcyclist scenarios — testing steering correction for the departing vehicle. It does not test AEB response for the receiving vehicle when an adjacent vehicle intrudes into its path. NHTSA FMVSS No. 127 [7] covers forward collisions only. UNECE Regulation No. 152 [9] similarly covers only forward collision scenarios. The receiving vehicle's AEB lateral response remains unvalidated in any current framework. This study provides a parameterised simulation foundation — speed 60–100 km/h, drift 0.3–1.2 m/s, gap 2.5–5.0 m — for developing such tests. ELK and AEB represent two sides of the same collision. This study addresses the side that remains unvalidated.

None of these directions should be implemented on this study alone. Real-world sensor validation, failure mode effects analysis, and ISO 21448 safety case development are required before any production system modification.

VI. Conclusion

Modern AEB systems have made significant progress in reducing forward collision fatalities. However, lateral intrusion scenarios where a vehicle departs an adjacent lane into the ego vehicle's path at highway speed—remain entirely absent from Euro NCAP active safety validation. Analysis of 1.54 million German crash records shows motorcycle leaving-carriageway events produce 44.1% severe injury rates, 2.37× the national baseline. This study tested whether production AEB architecture can handle this scenario. Across 100,000 Monte Carlo scenarios, validated across three independent random seeds (std $= 0.020\%$), production architecture produced collision outcomes in 99.57% of cases.

The sensors worked: all three production sensors detected the departing threat at $t = 5.011$ s, just 0.011 s after departure began. The problem was what came after. TTC-based AEB logic is built for vehicles approaching head-on at high closing speeds. In a lateral intrusion at 0.70 m/s drift, TTC declines 2.5× more slowly than in a standard forward collision. The activation threshold was not crossed until $t = 8.05$ s, leaving just 1.10 s before collision at $t = 9.15$ s—not enough time to stop at 80 km/h. Testing eight different TTC thresholds from 1.5 s to 5.0 s confirmed the same outcome every time. This is not a calibration problem. It is a structural one.

Counterfactual testing showed where improvement is possible. Upgrading side radar speed threshold while keeping 8 m range reduced collisions by only 2.19 pp — range is the real bottleneck. Adding corner radar (120° FOV, 50 m range) reduced collisions by 22.15 pp—10.1× larger improvement, because it addressed both coverage and range together. Removing all sensor limits entirely produced a floor of 46.62% — meaning 53.2% of failures can be addressed through better sensor architecture, while 46.6% reflect physics constraints no braking system can overcome at these speeds. Even the best tested actuation combination (0.5 s reaction, 10.0 $\mathrm{m/s}^2$) reached only 16.48% under perfect detection. The near-full-speed impacts that result — mean 22.77 m/s — directly explain the severe injury rates seen in Phase 1 crash data.

Four contributions emerge from this work. First, a simulation-based quantification of lateral detection gaps in production AEB through a validated 100,000-scenario Monte Carlo framework. Second, counterfactual isolation showing corner radar delivers 10.1× greater improvement than side radar threshold upgrades. Third, a 53.2% versus 46.6% failure attribution separating what sensor improvement can address from what physics constrains. Fourth, TTC-threshold-independence confirmed across the full realistic production range — the failure cannot be fixed by adjusting a single parameter.

Future work should focus on real-world sensor validation with production radar hardware, RCS-based detection modelling, curve geometry scenarios, variable drift profiles, and lateral evasive steering to address the physics-limited floor.

The parameterised simulation framework developed here—with validated kinematics, reproducible results, and a defined parameter space—provides a foundation for Euro NCAP lateral intrusion test design. Production AEB detects lateral threats. It does not act on them in time. Closing that gap requires both the hardware to see further and the decision logic to respond to what it already sees.

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